

Topic C10

Chemistry of the environment

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

C10.1	Water
C10.2	Fertilisers
C10.3	Air quality and pollutants
C10.4	Carbon dioxide and climate
C10.5	Environmental evaluation

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Water supply and treatment

Natural water may contain dissolved salts, microorganisms and suspended solids. Potable water is safe to drink, not chemically pure. Treatment commonly includes screening or sedimentation, filtration and chlorination. Distillation produces pure water but requires much energy. Tests for water include anhydrous copper(II) sulfate, white to blue, and cobalt(II) chloride paper, blue to pink.

Examiner insight

“Potable” means safe for consumption; dissolved minerals may remain.

2. Fertilisers

Plants require nitrogen for amino acids and proteins, phosphorus for roots and energy-transfer compounds, and potassium for enzyme function and healthy flowers and fruits. NPK fertilisers replace mineral ions removed from soil. Excess fertiliser can cause eutrophication: algal growth, light blocking, plant death, decomposition and oxygen depletion.

Examiner insight

Explain eutrophication as a linked sequence ending in death of aquatic organisms from low dissolved oxygen.

3. Air composition and pollutants

Dry clean air is about 78% nitrogen, 21% oxygen and small amounts of noble gases and carbon dioxide. Carbon monoxide from incomplete combustion is toxic because it reduces blood oxygen transport. Sulfur dioxide and nitrogen oxides cause acid rain; particulates damage lungs and contribute to global dimming. Catalytic converters reduce carbon monoxide, hydrocarbons and nitrogen oxides.



Examiner insight

Match each pollutant to its source, effect and control method.

4. Greenhouse gases and climate

Carbon dioxide, methane and water vapour absorb outgoing infrared radiation and re-emit it, warming the lower atmosphere. The natural greenhouse effect makes Earth habitable; increased greenhouse-gas concentrations enhance warming. Carbon dioxide sources include fossil-fuel combustion and deforestation; sinks include photosynthesis and ocean absorption.

Examiner insight

Do not say greenhouse gases “trap sunlight”; they absorb outgoing infrared radiation.

5. Evaluating environmental choices

Scientific evaluation weighs evidence, uncertainty and trade-offs. Reducing emissions can involve renewable energy, efficiency, public transport, carbon capture and reforestation. Life-cycle analysis considers raw materials, manufacture, use and disposal rather than only the point of use.

Examiner insight

A strong conclusion states a choice and justifies it using at least two relevant criteria.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

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Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Potable water is water that is [1]
 - A pure H₂O only
 - B safe to drink
 - C always distilled
 - D free of dissolved ions

2. Anhydrous copper(II) sulfate changes from [1]
 - A blue to white
 - B white to blue
 - C pink to blue
 - D green to red

3. Which element in fertiliser is needed for amino acids? [1]
 - A nitrogen
 - B argon
 - C chlorine
 - D neon

4. What begins eutrophication after fertiliser runoff? [1]
 - A algal bloom
 - B increased oxygen
 - C fish reproduction
 - D water distillation

5. What is the approximate oxygen percentage in dry air? [1]
 - A 1%
 - B 21%
 - C 50%
 - D 78%

6. Which pollutant reduces oxygen transport in blood? [1]
 - A carbon monoxide
 - B carbon dioxide
 - C nitrogen
 - D water vapour

7. Which pollutants cause acid rain? [1]
 - A oxygen and argon
 - B sulfur dioxide and nitrogen oxides
 - C methane and helium
 - D hydrogen and nitrogen

8. Particulates can cause [1]
- A respiratory damage
 - B more blood oxygen
 - C neutral rain
 - D ozone formation only
9. Greenhouse gases absorb mainly outgoing [1]
- A ultraviolet
 - B infrared
 - C radio only
 - D visible light only
10. Which process removes carbon dioxide from air? [1]
- A combustion
 - B respiration
 - C photosynthesis
 - D thermal decomposition

Section B · Structured questions

11 A town treats river water for drinking.

(a) State two possible contaminants. [2]

(b) Describe filtration and chlorination. [2]

(c) Explain why potable water is not necessarily pure. [2]

(d) Describe a chemical test for water. [2]

[Total: 8]

12 Fertiliser from farmland enters a lake.

(a) State the useful ions supplied by an NPK fertiliser. [2]

(b) Describe the growth of algae. [1]

(c) Explain the fall in dissolved oxygen. [3]

(d) Explain why fish may die.

[2]

[Total: 8]

13 A petrol engine burns fuel incompletely.

(a) Name two pollutants produced.

[2]

(b) Explain why carbon monoxide is harmful.

[2]

(c) State one effect of particulates.

[1]

(d) Explain how a catalytic converter reduces emissions.

[3]

[Total: 8]

14 The concentration of atmospheric carbon dioxide increases.

(a) State two human causes.

[2]

(b) Explain the greenhouse mechanism.

[3]

(c) State two possible climate effects. [2]

(d) State one carbon sink. [1]

[Total: 8]

15 A city compares electric buses with diesel buses.

(a) State two factors in a life-cycle assessment. [2]

(b) Give two advantages of electric buses in cities. [2]

(c) Give two possible disadvantages. [2]

(d) Write a justified conclusion. [2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 B. It is treated to be safe, though minerals can remain.
- 2 B. Hydration forms blue copper(II) sulfate.
- 3 A. Nitrogen is required to make amino acids and proteins.
- 4 A. Nutrients stimulate rapid algal growth.
- 5 B. Air contains about 21% oxygen.
- 6 A. Carbon monoxide binds strongly to haemoglobin.
- 7 B. Their acidic oxides form acids in atmospheric water.
- 8 A. Fine particles enter lungs.
- 9 B. Earth emits infrared radiation.
- 10 C. Plants incorporate carbon dioxide into biomass.

Solutions · Structured questions

11. 11(a) Microorganisms, suspended solids, dissolved salts or pollutants [any 2]. (b) Filtration removes insoluble particles; chlorine kills microorganisms [2]. (c) Safe dissolved ions may remain [2]. (d) Anhydrous CuSO_4 white to blue or cobalt chloride paper blue to pink [2].

12. 12(a) Nitrogen-containing ions, phosphate and potassium ions [2]. (b) Rapid algal bloom [1]. (c) Light is blocked, plants and algae die, and decomposers respire and use oxygen [3]. (d) Insufficient oxygen for aerobic respiration [2].

13. 13(a) Carbon monoxide, unburned hydrocarbons, particulates or nitrogen oxides [any 2]. (b) It binds haemoglobin and reduces oxygen transport [2]. (c) Respiratory disease or global dimming [1]. (d) It catalyses oxidation of carbon monoxide and hydrocarbons and reduction of nitrogen oxides to less harmful products [3].

14. 14(a) Fossil-fuel combustion and deforestation or cement manufacture [2]. (b) Carbon dioxide absorbs outgoing infrared and re-emits it, reducing heat loss [3]. (c) Temperature rise, ice melt, sea-level rise or altered weather [any 2]. (d) Photosynthesis, oceans or carbonate rocks [1].

15. 15(a) Raw materials, manufacture, transport, use, disposal or recycling [any 2]. (b) Lower point-of-use pollutants and noise, and possibly lower carbon dioxide [2]. (c) Battery materials, electricity source, cost or disposal [2]. (d) Valid choice linked to evidence and local electricity mix [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.